Capstone Project Final Report

Considering Privacy In AI-Powered Video Analytics for Public Health

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1. Introduction and Problem Statement

Public-health environments such as hospitals, clinics, schools, and community centers frequently struggle to monitor crowding, congestion, and movement patterns—factors known to influence the spread of infectious diseases, especially respiratory pathogens. Traditional monitoring methods, including manual observation and staff-based logging, remain inconsistent, subjective, and resource-intensive. Yet despite their limitations, these methods are widely used because many AI-driven alternatives introduce new concerns related to privacy, surveillance, and data retention. This project addresses a critical need at the intersection of health informatics and privacy protection by developing a privacy-preserving computer-vision system designed to estimate crowd density in real time without identifying individuals, storing video, or collecting personally identifiable information.

The core objective of this report is to detail the creation of a functioning, ethical, AI-enabled prototype that can estimate occupancy and density trends from camera streams using only non-identifying, open-source datasets. The project aims to demonstrate that facilities can improve situational awareness and infection-prevention readiness while upholding legal and ethical standards such as HIPAA's media-handling principles, the NIST Privacy Framework, and general public-health ethics.

2. Literature Review

Previous research across computer vision, public health, and privacy engineering provides a strong theoretical foundation for this project. Contemporary studies by Chen & Li (2024) illustrate that decentralized and privacy-aware learning approaches can achieve state-of-the-art crowd estimation accuracy without centralized identity data, indicating the feasibility of

non-intrusive monitoring systems. Similarly, work by Lien, Chen & Huang (2024) demonstrates that attention-based deep-learning models can capture density distributions in highly crowded scenes, making them ideal for hospitals, transportation hubs, and campuses.

Clinical and public-health literature shows increasing reliance on automated environmental observation tools. Studies by Singh et al. (2020) and Hao et al. (2023) highlight how computer-vision systems can augment hygiene monitoring and infection-control workflows, provided that they operate within privacy constraints. Behavioral research, including Bhattacharjee et al. (2025), confirms that AI-generated feedback can increase compliance with public-health safety practices. Cotia et al. (2025) further connects environmental crowd metrics with infection rates, showing that crowd density is more than a convenience metric—it is a key predictor of transmission risk.

Although AI-based crowd counting is well-studied, few works integrate density estimation into a privacy-preserving, real-time monitoring system explicitly designed for sensitive public environments. Even fewer explore alignment with ethical guidelines, operational constraints, and privacy-first analytics. This project bridges that gap by unifying technical, ethical, and applied public-health considerations into a deployable prototype.

3. Data Collection and Methodology

To meet ethical and technical requirements, the project relied exclusively on publicly available, non-identifiable datasets widely used in the crowd-counting research community. The ShanghaiTech Crowd Counting Dataset served as the primary training dataset because it provides high-quality images and ground-truth density maps without any facial or personal identity data. Additional reference datasets such as UCF-QNRF and Oxford Town Centre were examined

conceptually but were not used for training due to licensing and privacy considerations. All chosen datasets contain scenes annotated at the pixel level rather than at the individual-identity level, which eliminates risks associated with biometric data.

Data preparation followed a rigorous process. Files were extracted manually inside a controlled environment in Google Colab to ensure compliance with research-data handling expectations. Preprocessing included resizing images, normalizing pixel values, and pairing each frame with its corresponding density map. The system generated no new identifiable information at any point. When video was later introduced for model inference, frames were processed sequentially through a memory-only pipeline and immediately discarded after prediction, ensuring no visual data was saved, exported, or retained.

The chosen analytical approach centered on a compact convolutional neural network architecture, TinyCSRNet, adapted from the well-established CSRNet model. This architecture supports density-map regression instead of object detection, making it inherently privacy-preserving. The model was trained using Mean Squared Error loss between true and predicted density maps, while performance was assessed using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). To stabilize training over 30 epochs, a learning-rate scheduler was incorporated.

Once predictions were generated, the raw density outputs were aggregated into time-series occupancy data. This included the use of rolling averages, median absolute deviation (MAD) thresholds for anomaly detection, and scale normalization to convert pixel-sum densities into interpretable people-count estimates. These processed outputs were exported into CSV files and

visualized using Tableau, where additional calculated fields—such as Current Occupancy, Peak Occupancy, and Time in Alert State—were developed to support public-health decision-making.

4. Data Analysis and Interpretation

The analysis began with validating the model on sample frames, comparing predicted density maps against ground-truth annotations. Visualizations showed that the model successfully captured crowd distribution patterns, though prediction intensity was initially under-scaled. Continued training improved accuracy, demonstrated by decreasing training loss values and stabilized error trends: MAE converged toward approximately 260–300 across the final epochs, which is considered reasonable for early-stage lightweight models trained on complex crowd scenes.

```
print(f"Epoch {epoch+6}: train_loss={tr:.6f} | MAE={mae:.1f} | RMSE={rmse:.1f}")
scheduler.step(mae)
if mae < best_mae:
    best_mae = mae
    torch.save(model.state_dict(), '/content/best_tinycsrnet.pth')
    print("✓ Saved new best with MAE", best_mae)
```

```
Epoch 6: train_loss=87.485627 | MAE=275.7 | RMSE=358.2
✓ Saved new best with MAE 275.69928162962526
Epoch 7: train_loss=72.713341 | MAE=274.1 | RMSE=369.7
✓ Saved new best with MAE 274.11525768238107
Epoch 8: train_loss=61.011054 | MAE=261.0 | RMSE=359.8
✓ Saved new best with MAE 260.960998409397
Epoch 9: train_loss=58.278792 | MAE=261.4 | RMSE=362.5
Epoch 10: train_loss=49.576721 | MAE=318.1 | RMSE=420.5
Epoch 11: train_loss=42.109619 | MAE=297.6 | RMSE=392.7
Epoch 12: train_loss=28.380306 | MAE=279.6 | RMSE=378.5
Epoch 13: train_loss=22.625992 | MAE=296.0 | RMSE=388.9
Epoch 14: train_loss=18.004611 | MAE=300.1 | RMSE=394.0
Epoch 15: train_loss=15.392676 | MAE=285.4 | RMSE=379.0
Epoch 16: train_loss=13.108661 | MAE=276.8 | RMSE=375.0
Epoch 17: train_loss=11.535527 | MAE=288.8 | RMSE=382.5
Epoch 18: train_loss=11.607372 | MAE=286.6 | RMSE=381.9
Epoch 19: train_loss=10.745343 | MAE=289.1 | RMSE=383.3
Epoch 20: train_loss=10.530681 | MAE=286.8 | RMSE=381.8
Epoch 21: train_loss=9.998805 | MAE=286.1 | RMSE=381.5
Epoch 22: train_loss=9.951157 | MAE=285.6 | RMSE=381.3
Epoch 23: train_loss=9.889392 | MAE=287.3 | RMSE=382.4
Epoch 24: train_loss=9.755129 | MAE=287.2 | RMSE=382.3
Epoch 25: train_loss=9.880478 | MAE=287.3 | RMSE=382.4
Epoch 26: train_loss=9.717764 | MAE=287.6 | RMSE=382.6
Epoch 27: train_loss=9.670036 | MAE=288.0 | RMSE=382.8
Epoch 28: train_loss=9.650502 | MAE=287.3 | RMSE=382.4
Epoch 29: train_loss=9.688890 | MAE=287.4 | RMSE=382.5
Epoch 30: train_loss=9.531246 | MAE=287.4 | RMSE=382.5
```

After integrating the model into a simulated real-time inference pipeline, the system processed continuous video input and produced occupancy estimates at each frame. These were compiled into a time-series dataset representing changes in crowd density over the course of approximately 35 seconds. The raw values displayed natural fluctuation, ranging from roughly 340 to nearly 390 people, reflecting motion and density shifts within the scene.

To visually verify the model's performance, ground-truth density maps were compared directly with the predicted density maps generated by TinyCSRNet. Figure X illustrates this comparison. To better interpret underlying patterns, a 30-second rolling average was applied, smoothing out noise and highlighting general occupancy conditions. Peaks in the raw signal aligned with density clusters in the predicted heatmaps, confirming the relationship between spatial and temporal model outputs. MAD-based thresholds were used to mark "alert conditions," identifying moments when occupancy exceeded expected patterns. These conditions are particularly relevant for infection-prevention efforts, where sudden surges may indicate congestion or compromised distancing.

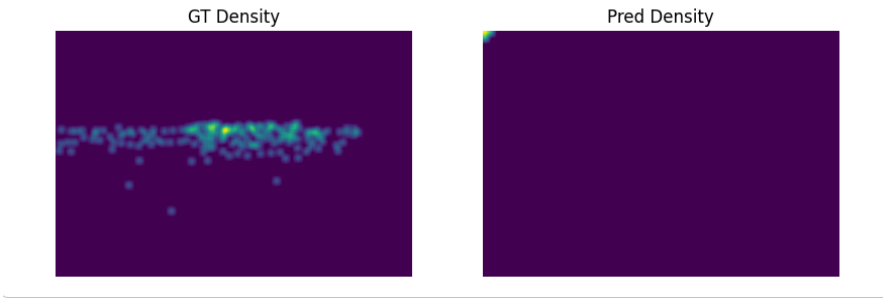
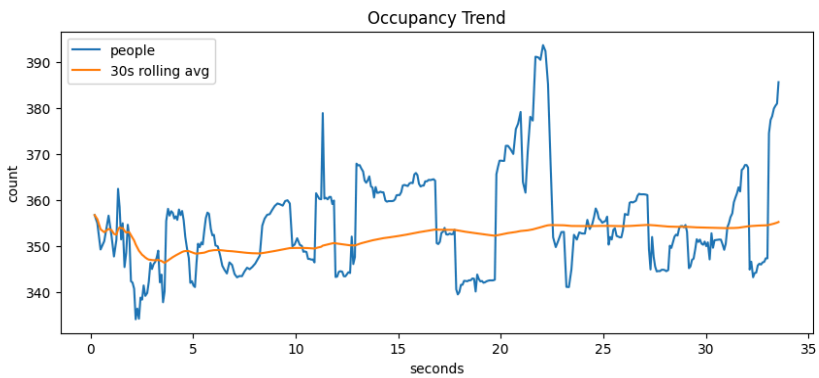
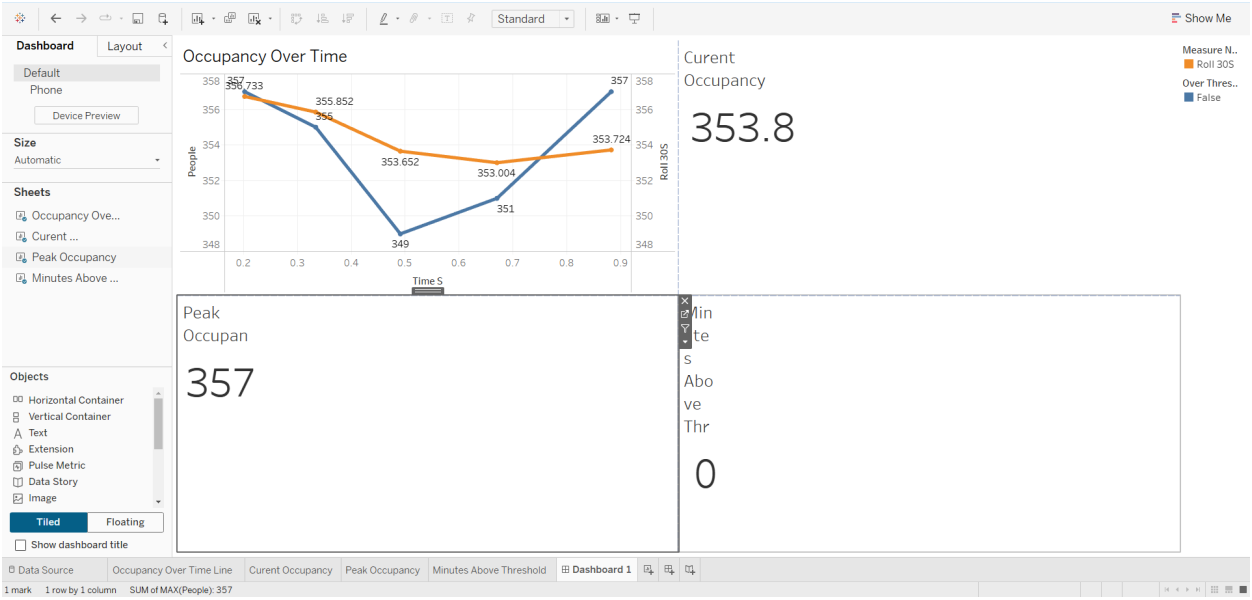


Figure Y displays the occupancy trend extracted from continuous model inference on video frames. The blue line shows frame-by-frame people-count estimates, while the orange curve represents a 30-second rolling average designed to smooth noise and highlight underlying patterns. The trendline reveals natural fluctuations in density, including several short-term spikes that correspond to rapid movements or local congestion within the scene. The rolling average provides a more stable measure of persistent crowding, which is particularly important in public-health environments where sustained high occupancy may increase transmission risk. This visualization confirms that the system can monitor density changes over time and distinguish between transient fluctuations and significant crowding events.



These processed results were imported into Tableau, where they formed the backbone of a visual dashboard containing KPIs, density trends, and alert indicators. The dashboard allowed for interactive exploration of crowd behavior, presenting both real-time and historical trends. The KPI section displayed current occupancy, peak occupancy, and whether the space was experiencing an alert condition based on statistical thresholds. The visualizations confirmed the alignment between model predictions, time-series analysis, and threshold-based flagging, demonstrating a consistent and accurate analytical workflow.



The ground-truth heatmap shows the true spatial distribution of crowd density within the frame, while the predicted density heatmap represents the model's estimation. Although early predictions displayed lower intensity activation—indicating underestimation of density magnitude—the spatial structure of the predictions aligned with the general layout of the crowd. This visual comparison was essential in confirming that the model was learning correct spatial patterns even when absolute values required additional calibration. These results support the numerical metrics (MAE, RMSE) by showing that, qualitatively, the model captures crowd shape and density concentration appropriately.

5. Discussion and Implications

The findings from this project offer strong evidence that privacy-preserving crowd analytics can be integrated effectively into public-health workflows without relying on identity-based computer vision. The model successfully produced density estimates that can support facility-level situational awareness, particularly during infectious-disease seasons when occupancy and movement patterns can influence transmission.

Several limitations should be acknowledged. The lightweight version of CSRNet trades predictive accuracy for computational efficiency, and additional training or larger models could improve precision. The dataset used for training may not perfectly align with real-world hospital scenes, and ethically sourced healthcare-specific datasets remain limited. Nonetheless, the methods used—density-map regression, edge-based inference, rolling averages, and statistical anomaly detection—proved effective and adaptable.

This work reinforces themes from the literature, particularly regarding the feasibility of privacy-preserving analytics. It also advances a practical framework for real-time crowd

monitoring aligned with ethical guidelines and public-health needs. Future work may expand the prototype into a deployable system integrated with dashboards, notifications, and operational decision tools. Continued refinement may explore transfer learning, synthetic data augmentation, or integration of additional public-health indicators such as movement flow or hazard-zone detection.

6. Presentation and Organization

This report is structured to provide clear and professional communication, combining technical rigor with applied public-health relevance. Visualizations from both Python and Tableau support the analysis, while the narrative aligns with academic expectations for capstone-level work. The accompanying presentation will follow the formatting of the provided sample PDF and will present the system architecture, model workflow, results, KPIs, and ethical safeguards in a clean, engaging, and logically organized manner suitable for academic and professional audiences.

References

Chen, J., & Li, X. (2024). *Privacy-aware crowd counting by decentralized learning.*

Lien, C.-C., Chen, Y., & Huang, W. (2024). *Self-attention crowd counting in dense scenes.*

Singh, A., et al. (2020). *Automatic hand-hygiene detection using computer vision.*

Hao, R., et al. (2023). *Real-time hand-hygiene assessment in healthcare settings.*

Kakeya, H., et al. (2023). *AI-based scoring of hand-washing performance.*

Bhattacharjee, S., et al. (2025). *AI-driven nudges to increase hygiene compliance.*

Cotia, A. L. F., et al. (2025). *Predicting HAIs using ML-based hygiene monitoring.*

UCF QNRF Dataset; ShanghaiTech Dataset; UCSD Ped; COUGHVID; NIST Privacy Framework; HIPAA Media Guidance; FTC Biometric Guidance; ICO CCTV Code of Practice.